

Software Requirements Specification

On

LIBRARY MANAGEMENT **SYSTEM**

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1. INTRODUCTION

1.1 Purpose

The purpose of this document is to specify the complete functional and non-functional requirements for the web-based Library Management System (LMS). This document serves as the structural blueprint for developers, QA testers, and stakeholders during implementation.

1.2 Scope

The Library Management System is a centralized Web Application designed to replace manual, paper-based ledger systems in educational libraries.

- **Core Mission:** Act as a digital bridge connecting students, library operational staff, and management.
- **Key Mechanisms:** Real-time catalog tracking, automated algorithmic fine calculation, and instant digital receipt generation.
- **Boundaries:** This system handles digital records and transaction states. Physical book tracking is handled via barcode scan inputs.

1.3 References

- *IEEE Std 830-1998:* IEEE Recommended Practice for Software Requirements Specifications.
- *Software Engineering: A Practitioner's Approach* by Roger S. Pressman.

2. FUNCTIONAL REQUIREMENTS

Functional requirements define the specific tasks, operations, and behaviors that the software must perform.

2.1 User Authentication and Profile Management

- **Registration:** The system shall allow students and staff to register by providing unique identification details, contact numbers, and email addresses.
- **Secure Login:** Users must authenticate using a unique Username/ID and password. The system must restrict dashboard access based on the authorized user role (Student, Librarian, Management).
- **Profile Maintenance:** Users must be able to view and update their profile details, including contact information and passwords.

2.2 Catalog Search and Book Discovery

- **Search Functionality:** The system shall provide a search bar for users to find books by Title, Author, Subject, ISBN, or Keywords.

- **Availability Status:** The catalog must display real-time availability details for each book (e.g., "Available", "Issued", "Reserved", "Lost").

2.3 Book Circulation Operations

- **Book Issue:** Librarians must be able to issue a book to a registered student by scanning or inputting the student ID and the book's unique barcode.
- **Book Return:** Librarians must be able to process returned books, updating the book status back to "Available" and logging the return date.
- **Reservations:** If a desired book is currently issued out, students shall have the option to reserve it. The system will notify the student once the book becomes available.

2.4 Fine Calculation and Receipt Generation

- **Automated Fine Assessment:** The system shall automatically calculate overdue fines based on the due date and a predefined daily rate.
- **Receipt Generation:** Upon fine payment or transaction completion, the system must generate a digital receipt containing the transaction ID, student details, book details, and payment breakdown.

2.5 Inventory and Catalog Management

- **Add/Update/Remove Titles:** Librarians must be able to add new books to the inventory, edit existing book metadata, or remove damaged/lost copies from the active catalog.
- **Category Tagging:** Books must be organized and filtered by genres, departments, or reference sections.

2.6 Reporting and Analytics

- **Circulation Reports:** The system shall generate weekly and monthly reports detailing books issued, returned, and overdue.
- **Financial Reports:** Management staff must be able to generate financial summaries tracking the total fines collected and outstanding dues.

3. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements specify the quality attributes, operational constraints, and performance benchmarks of the software.

3.1 Performance

- **Response Time:** The system must process search queries and load pages within 2 seconds under normal network conditions.
- **Concurrency:** The system must support at least 500 concurrent active sessions without degradation in performance or data processing bottlenecks.

3.2 Security

- **Data Encryption:** All sensitive user data, including authentication passwords, must be encrypted using industry-standard hashing algorithms (e.g., bcrypt or SHA-256) before database storage.
- **Session Management:** User sessions must automatically timeout after 15 minutes of inactivity to protect unauthorized access on public terminals.

3.3 Reliability and Availability

- **Uptime:** The web application shall maintain an operational availability of 99.5% excluding scheduled maintenance windows.
- **Data Backup:** The system must execute automated daily incremental backups and weekly full backups of the database to prevent permanent data loss.

3.4 Usability

- **User Interface:** The user interface must be clean, accessible, intuitive, and fully responsive across standard devices, including desktops, tablets, and smartphones.
- **Error Handling:** The system must display clear, user-friendly error messages instead of system stack traces when an invalid operation occurs.

4. FEASIBILITY STUDY

4.1 Technical Feasibility

The technologies required to build the web application—such as standard modern web frameworks (React/HTML5 for frontend, Node.js/Django for backend) and relational database management systems (MySQL or PostgreSQL)—are mature, widely documented, and readily accessible. The development team possesses the necessary programming proficiency to build, test, and deploy the application securely.

4.2 Economic Feasibility

The financial investment required to develop this system is minimal compared to the long-term operational savings. By moving away from manual ledger tracking, the organization will drastically reduce expenses on physical stationery, labor costs, and lost inventory. Open-source technologies can be utilized to eliminate software licensing fees, ensuring a high return on investment.

4.3 Operational Feasibility

The current library workflows map seamlessly into the automated modules designed for this system. Librarians and management staff will experience a reduction in repetitive administrative burdens, which ensures strong organizational acceptance. Minimal training will be required for staff to master the dashboard due to its emphasis on a clean, intuitive layout.